INVENTION DISCLOSURE FORM

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1. **Particulars of Inventors**

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2. **Provide a brief descriptive title of the invention:**
System and Method for Constraint-Based Multi-Objective Optimization of M&A Deal Structures Using Pareto Efficiency and AI-Assisted Decision Support

3. **In 100 words or less, please provide an abstract or summary of the invention:**
A mixed-integer constrained multi-criteria decision support framework for constructing mergers/acquisitions (M&As) has been developed. The framework represents M&A structures using variables, such as the cash/stock ratio and estimated synergies, and assesses the value of those structures by calculating expected post-acquisition EPS using a deterministic financial analysis. Using a grid search function, multiple possibilities are produced from which to develop many possible M&A structures by being filtered through rigorous financial constraints and then optimized using Pareto Efficiency to create the best trade-off between expected returns (i.e., EPS) and the level of risk (i.e., sensitivity of projected synergies). The best structures then go through an artificial intelligence based interpretation module for generating structured strategic insights from the M&A structured information and producing structured strategic insights. Consequently, the M&A framework allows for systematic, repeatable, and data-driven evaluations of M&A structures.

4. **Detail description of the invention:(Answer to all below are required in detail)**

    a. **Problem the invention is solving**

Current methods for making decisions regarding mergers and acquisitions (M&A) are reliant on static, spreadsheet-based methods for analyzing transactions based on financial models that can only provide a limited view of possible outcomes based on pre-defined criteria. Current methods are not capable of dynamically generating an expansive number of alternative deal configurations depending on cash-stock mix and synergies. Current decision-making methods also do not assign explicit values related to the trade-offs between potential returns (as indicated by post-acquisition earnings per share, or EPS) and the associated uncertainty in realization of projected synergies. Therefore, the likelihood that executives will choose a deal structure that is not the best available or can be mathematically dominated by better alternatives is much higher because they do not have proper knowledge of higher-performing alternatives. Currently, there is no structured framework available to provide a systematic process through which to identify and compare efficient deal configurations based on financial constraints and multi-objective optimization.

b. General Utility/application of the invention

The invention is intended for providing assistance in the evaluations and structuring of M&A deals by Investment Analysts, Corporate Finance Departments, and Strategic Planning Divisions through use of computerized financial models. The invention allows for the evaluation of multiple variations of the financing combination (use of cash and/or stock) and the effect on post M&A EPS and risk associated with that combination. The invention is particularly beneficial in assessing accretion vs. dilution analysis, where management evaluates the effect of a proposed transaction on the value of the shareholders. Furthermore, the invention may be useful for academics and other researchers studying optimization methods of financial modeling. As a means to systematically search for feasible deal configurations and to present optimal trade-off analysis, the invention serves as a tool that permits informed, data-driven decision-making in the complex environment of M&A.

c. Advantages of the invention disclosing about the increased efficiency/efficacy

The invention offers a number of advantages over traditional M&A assessment methods. Instead of using a conventional spreadsheet model that evaluates only a very small number of pre-defined scenarios, the inventive system performs an extensive evaluation of many different possible deal configurations through a structured grid search method. The inventive system uses multi-objective optimization to support simultaneous evaluation of the conflicting objectives of maximizing post-deal EPS as well as minimizing the sensitivity of EPS to variations in synergy. The system uses a Pareto efficient solution to ensure that the decision maker is only presented with the non-dominated, optimal trade-off solutions; thus, allowing the decision maker to make decisions based on several different reasonable alternatives rather than just one possible outcome. Also, the constraint-based filtering ensures that the evaluated scenarios are all financially feasible and eliminates any of the high-risk or infeasible scenarios early on in the M&A evaluation process. The deterministic computation model provides for consistent and reproducible analyses across different scenarios. Furthermore, the incorporation of an AI-based interpretation layer improves the usability of the system by translating quantitative results into structured strategic insights, thereby reducing the cognitive complexity of the decision-making process.

d. Best way of using the invention as well as possible variants

Using the invention effectively requires an integrated computational system made up of the following three components: an input interface; optimization engine; and output visualization module. An end user enters key financial information related to the acquisition - i.e., acquirer and target EPS; total value of the transaction; and projected synergies into the input interface. The optimization engine then processes these financial inputs according to a deterministic financial model - thus producing multiple deal scenarios based on variations of cash and stock proportions via a pre-defined grid search method. After the model produces the various deal scenarios, each of the scenarios produced by the model is subsequently analyzed against pre-defined financial constraints (i.e. non-compliant scenarios are eliminated). The remaining compliant scenarios are subjected to a further refinement process based on the concept of Pareto efficiency in order to help the user identify optimum trade-offs between return (post-deal EPS) and risk (i.e. sensitivity to changes in projected synergy). The remaining compliant and refined scenarios are visually presented to the user via an appropriate means (e.g., risk-return scatter plot) and then processed by an artificial

intelligence-based interpretive module so that structured strategic insights can be generated for decision-making purposes.
The invention may be modified by altering the grid search resolution to allow a more fine or coarse decision-space search, including additional financial inputs such as debt financing, taxation, etc., and expanding optimization objectives in addition to EPS such as return on investment and/or valuation multiples. The invention can be made as a stand-alone desktop application, a web-based application, or integrated with existing financial-modeling tools. Also, the AI interpretation layer may be replaced or augmented with rule-based analytical modules based on where the invention will be deployed.

e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

The structured decision-support mechanism is seen as an effective pipeline for decision-making through the development of a series of financial inputs, exploring the alternative structures in a systematic manner, evaluating those structures based on constraints that have been established, and reporting the optimal combinations along with an explanation of the findings. The following stages define how the system will operate:

Stage 1 - Definition of Input
This stage begins with the user entering the minimum input financial data needed to evaluate each merger and acquisition. The primary inputs required to evaluate the merger or acquisition are as follows: acquirer's EPS; target's EPS; total value of the merger or acquisition; and expected synergy. The system also clearly defines the decision space based on the proportion of cash to stock used for the merger or acquisition, which typically has a range from 0% to 100% for each combination and is measured in discrete percentages.
At this point in the process, the system does not make any assumptions regarding the optimal structure for the merger or acquisition and instead generates a bounded search space based on reasonable financial input.

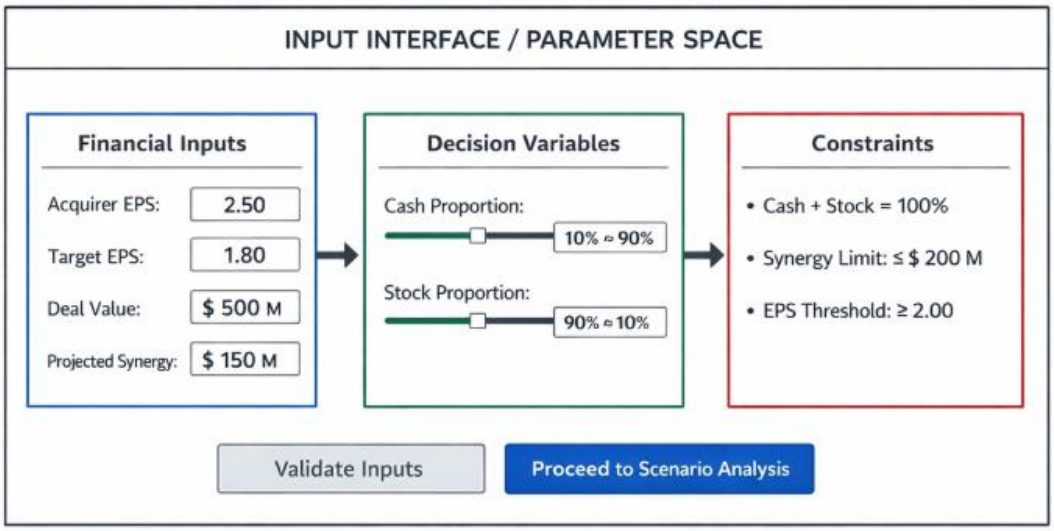

Fig. 1: Input Interface / Parameter Space

Stage 2 - Scenario Development (Exploration Phase)
Instead of evaluating only one potential merger or acquisition structure, the system generates several potential merger and acquisition structures based on the different possible combinations of cash and stock ratios respectively and the levels of expected synergy associated with each structure. This is accomplished by employing a discrete grid search methodology wherein the proportion of cash to

stock is varied incrementally from 0% to 100% and at each level of expected synergy (low, base and high).

Each of these different combinations of cash and stock represents a potential scenario for the merger or acquisition. Stage 2 of the process assures that no intuitive or pre-conceived assumptions regarding the structure of the merger or acquisition are made; instead, the full decision space is examined in a systematic process.

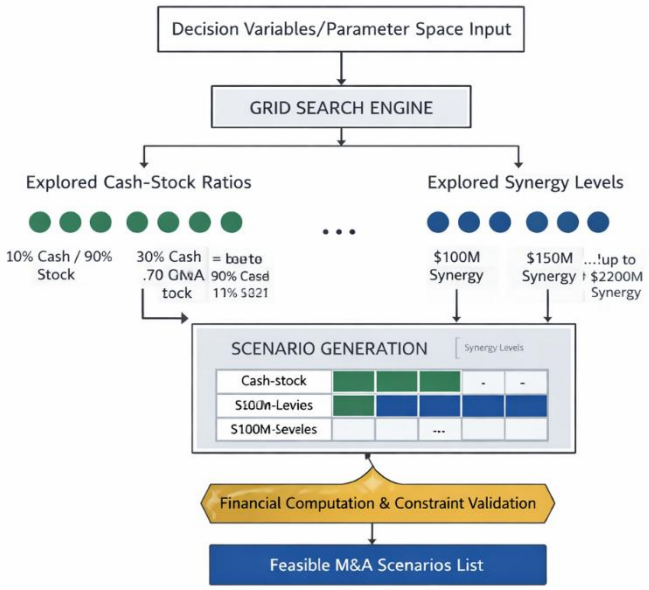

Fig. 2: Grid Search / Scenario Generation Flow

Stage 3: Financial Assessment of Each Scenario

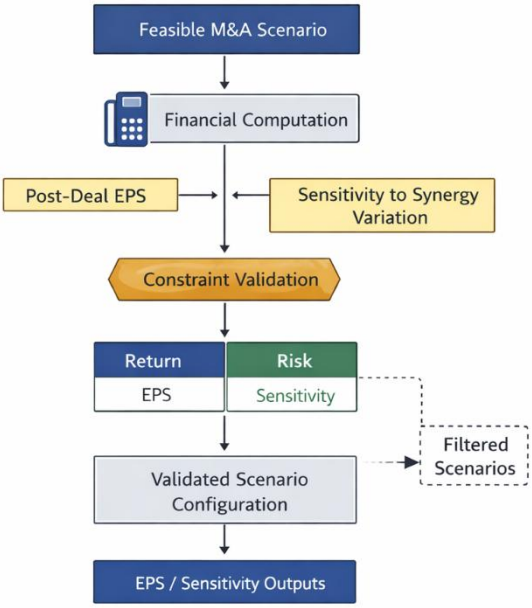

Fig. 3: Financial Model Flow Diagram

The system utilizes a deterministic financial model to determine the financial impacts of the deal after the two parties have merged, incorporating all financial data from both companies in order to compute a post-deal EPS based on the adjusted total number of shares issued.
This model operates with standardized assumptions, which allows for consistent evaluations across all scenarios. Therefore, the results of each scenario will be easily duplicated and compared.

Stage 4: Constraint-Based Filtering (Feasibility Screening)
Once all financial analyses have been performed, the system reviews each scenario for potential in the context of predetermined constraints (e.g., total financing must equal 100%, assumption of unrealistically high synergies is prohibited, and minimum acceptable EPS threshold must be met to prevent dilution). Any scenario that does not meet any of these criteria will be eliminated immediately and will not factor into the decision-making process.

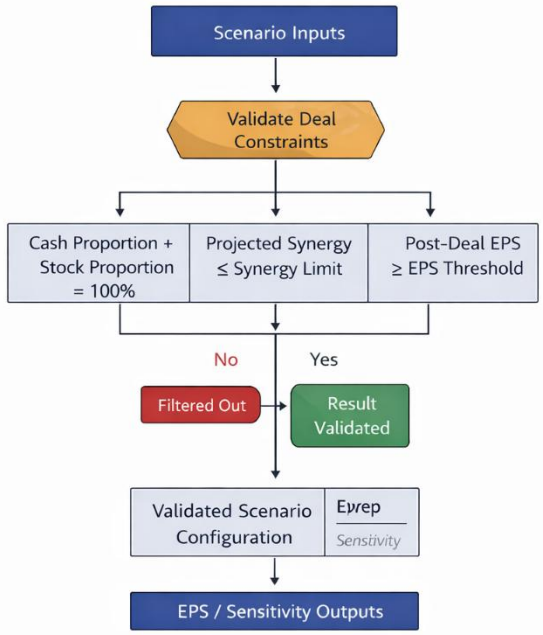

Fig. 4: Constraint Validation Process

Stage 5: Risk Quantification via Sensitivity Analysis
For all acceptable scenarios, the system will assess risk by measuring how sensitive the EPS after the deal is to changes in the estimated synergies associated with the two merging companies. The sensitivity is measured using a standardized sensitivity metric, which measures how much the EPS may differ from the projected EPS based on the realization of estimated synergies.
This analysis will introduce a measure of the uncertainty surrounding returns as well as measures of the overall stability of the returns.

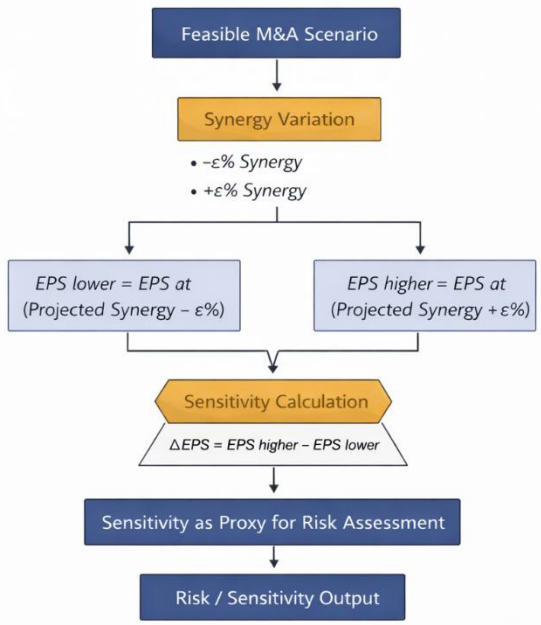

Fig. 5: Sensitivity / Risk Evaluation Flow

Stage 6: Pareto Optimization (Refining Solutions)

Next, all valid scenarios will be compared to determine which of them are Pareto efficient. For example, if there is a scenario that has higher expected return (EPS) and a lower chance of loss, it will be eliminated from consideration (as will any other scenario that does not meet this requirement). This filtering process results in many different optimal configurations, which are known to provide a variety of return-risk trade-offs; thus, they do not limit one to considering a "single-best" alternative.

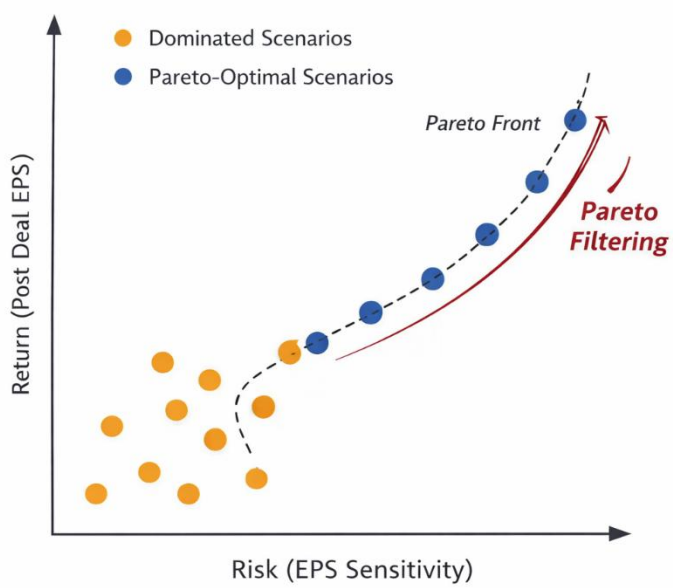

Fig. 6: Pareto Filtering Logic

Stage 7: Visualizing Trade-off Results

The results show optimal (lowest risk/highest return) alternatives plotted on a two-dimensional chart. The X-axis represents risk (sensitivity) and the Y-axis represents expected return (EPS). The clarity of this data representation helps demonstrate how increasing returns can increase/ decrease the amount of risk, and vice-versa; thus, facilitating an intuitive understanding of trade-off principles.

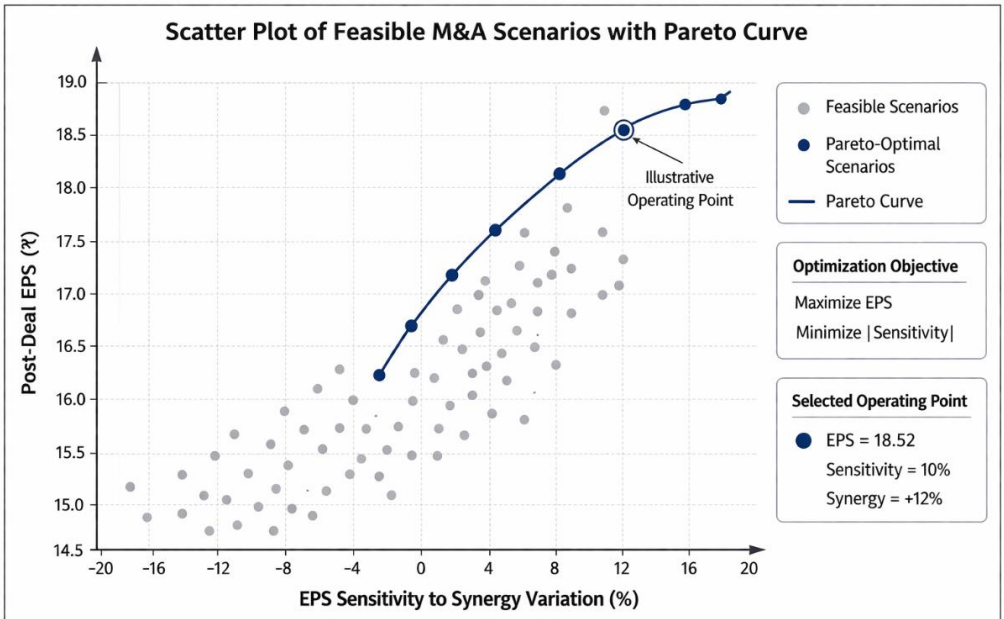

Fig. 7: Scatter Plot with Pareto Curve

Stage 8: AI-based Results Interpretation

The selected optimal scenarios are converted to a structured data format and sent to an AI data interpretation module. This module, does not create new data but only interprets already existing data to produce concise strategic explanations for each of the optimal scenarios. The interpretations will mainly focus on the reason for the structure of the deal providing high returns, an acceptable trade-off between risk and return, and an orientation towards a stable decision.

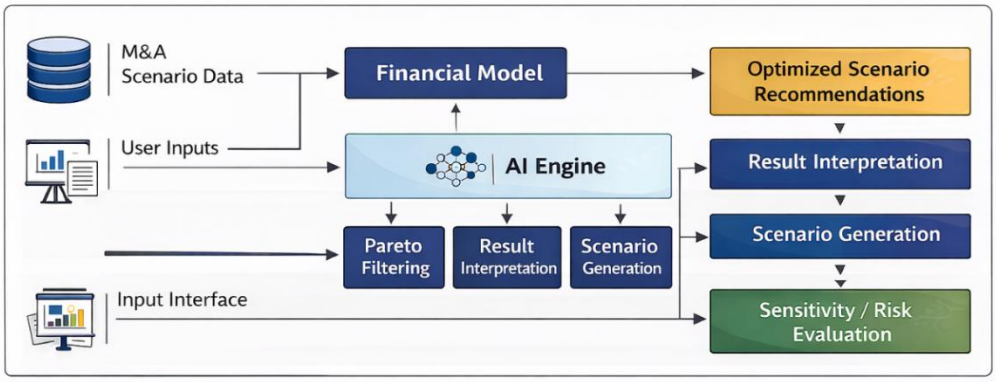

Fig. 8: AI Pipeline Diagram

Stage 9: Delivery of Final Output

The final output will be provided to the user using structured decision support information. This information will include recommended Deal Configuration, EPS values associated with those configurations, risk levels of those configurations, and a written description of each configuration and its importance.

The purpose of providing this information is to allow decision makers the opportunity to evaluate several feasible strategies and choose one best suited to their risk tolerance and financial objectives.

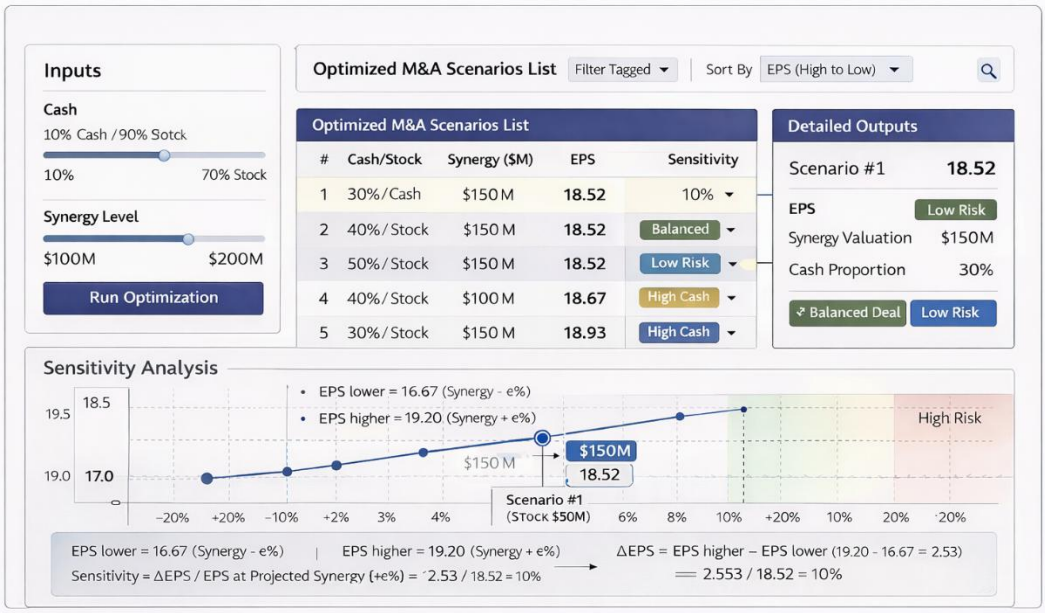

Fig. 9: Output UI Representation

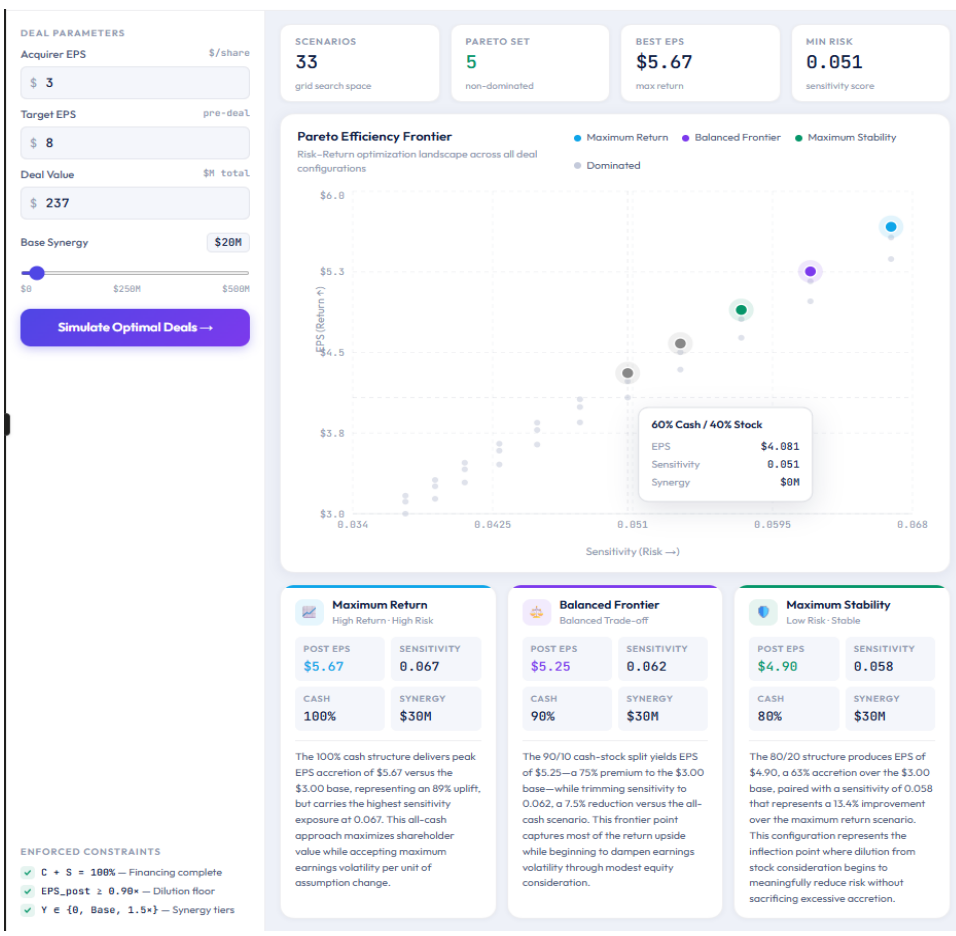

Fig. 10 Website Dashboard

5. **Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report)**
   No

6. **Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)**
   Current methods of making M&A decisions are mostly based on financial modeling, such as DCF, accretion/dilution, and comparable transactions. Financial analysts, investment bankers, and corporate finance teams typically use financial modeling to assess the financial impacts of proposed transactions through spreadsheet-based tools.
   Today's M&A modelling typically considers synergy estimates, financing options, and whether the transaction would be financially advantageous. The use of scenario analysis and sensitivity analysis also allows for understanding the potential impact of variation in different scenarios. But most of the analyses are done by running a limited number of manual scenarios instead of systematically exploring all possible decision points.
   Recently, machine learning and artificial intelligence have been integrated into various parts of the M&A process (such as identifying target companies and conducting due diligence). However, most of these advances have been focused on using predictive analytics, rather than structuring deals through a structured optimization process.
   All current systems have an ability to provide strong financial insight into a transaction, but most of them do not provide an integrated environment for decision support regarding M&A transaction structuring.

7. **List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.**
   There are many different methods currently used for helping with the decision-making process involved in making an acquisition or merger such as traditional financial modeling, scenario creation and analysis, and AI-based tools, but they all have limitations, such as limited ability to scale, optimize and provide structured decision-making support. The table below provides a comparison of each of the above methodologies along with the proposed invention.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Spreadsheet-based financial modeling (DCF, Excel models)	Evaluates only a limited number of manually defined scenarios and depends heavily on user assumptions	Uses automated grid search to systematically explore a wide range of possible deal configurations
2	Accretion/Dilution analysis models	Focuses only on EPS impact and does not account for risk or sensitivity to changing assumptions	Incorporates sensitivity analysis to quantify risk alongside return
3	Scenario-based financial evaluation	Scenarios are manually selected and may miss optimal or better-performing configurations	Generates all feasible scenarios and ensures comprehensive coverage of the decision space
4	Traditional decision-making approaches in Mergers & Acquisition	Relies on analyst intuition and static comparisons without structured optimization	Applies multi-objective optimization to evaluate trade-offs between return and risk
5	Basic financial modeling tools	Do not filter infeasible or impractical deal structures effectively	Uses constraint-based validation to eliminate unrealistic or invalid scenarios early
6	Existing AI applications in M&A (prediction, due	Focus mainly on prediction or data analysis, not on structured decision	Integrates AI as an interpretation layer on top of mathematically

	diligence tools)	optimization	verified optimal solutions

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

The suggested invention consists of several different technical components that work together to allow structured optimization of the M&A deal structure for decision support and optimization. The technical attributes, capabilities, and features follow:

1. Input Parameter Module

This module captures and validates the financial inputs used for analysis. Examples of inputs tracked in the Input Parameter Module include variables like the acquirer's and target's EPS, the acquired company's deal value, and the projected synergy. This module also establishes an acceptable range of cash and stock proportions to use as the basis for the decision-making process. This module will validate that the specified inputs are within acceptable financial bounds prior to their processing.

2. Scenario Generation Engine

The invention has a scenario generation engine that creates potential deal structures according to a systematic procedure. The scenario generation engine uses a discrete grid-based search that iterates through the various combinations of cash as a percentage of the stock and synergy levels specified for an M&A transaction. Therefore, rather than relying on manually selected transactions and entering these transactions, this engine will ensure that the exploration of the decision space is structured.

3. Deterministic Financial Computation Module

The deterministic financial computation module will perform all financial computations of each of the scenarios that the scenario generation engine has generated. All calculations will be based on the determination of the combined net income of both the acquirer and the acquiree after the transactions have been completed. The calculations will also adjust the acquirer's total number of shares issued and outstanding based on the amount of stock issued to both the acquirer and the acquiree and then derive the post-deal net income per share for the acquirer. The deterministic model used in the computations will also ensure that the results will be the same for identical input values.

4. The Constraint Validation Component

The system uses a financial rule set to impose restraints by making sure there is a complete financing package (cash + stock = 100%), using realistic synergy limitations, and enforcing a minimum acceptable EPS. The constraint validation module will then remove any scenarios that cannot be financed from further consideration.

5. The Risk Analysis Module (Sensitivity Analysis)

This module determines how each scenario's risk is affected by changes in synergy and how much the post-deal EPS will vary given different synergy values. This provides an estimate of how changes to an input will affect an output, allowing for an assessment of stability in addition to return.

6. The Multi-Objective Optimization Engine

The core technical element of the invention is an optimization engine for multi-objective optimization through simultaneous evaluation of all criteria. This allows for the balancing of return (EPS) and risk (sensitivity), without reducing to one single objective, therefore facilitating the structured analysis of trade-offs.

7. The Pareto Frontier Extraction Module

This module extracts optimal scenarios from all valid scenarios according to Pareto efficiency principles (comparing, eliminating dominated) in order to preserve the set of scenarios that is the best trade-off on the basis of competing objectives.

8. Graphical Representation of Results and Decision Support

The system provides the ability to visualize optimally configured products on a graphical level as well as provide interpretability to those optimally configured products through a combination of graphical formats

(e.g., risk vs. return scatter plot) and structured output format that highlights the most optimal configurations available.

9. Artificial Intelligence Interpretation
An AI-based interpretation module that converts structured numerical output into brief summary descriptions that help users understand what the implications are for each of the different deal configurations was built into the system. The AI-based interpretation module will also allow users to understand the different project configurations without affecting the mathematical computations performed by the system.

10. Integrated System
The modules comprise an integrated collaborative system, and all outputs from one module are inputs to another module. The integrated system architecture allows for seamless movement from input to optimized output, providing a logically coherent and consistent flow throughout the processing of input through the ultimate output of an optimized decision.

9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

The invention consists of several distinctive elements that distinguish it from the current methods used for evaluating mergers and acquisitions (M&A). The main novelty is the integration of constraint-based financial validation with multi-objective optimization in order to analyze structured deals. Traditional methods assess discrete scenarios independently; however, the present invention generates and assesses a full complement of feasible configurations for any given deal on a systematic basis via a discrete search process.

A principal novel feature is that both return and risk are taken into account simultaneously via the combination of Post-Deal Earnings Per Share (EPS) and a measure of quantified sensitivity. As such, trade-offs between profit and stability can be explicitly modelled, an issue that is not addressed directly by traditional accretion/dilution or Excel spreadsheets.

Another distinct feature is the implementation of Pareto efficiency to establish filtering criteria to produce a list of optimal deal structure options. Rather than producing one recommended solution, the proposed invention produces a list of non-dominated options for decision makers to select from depending on their individual risk tolerances.

The use of an AI-based layer separating the computation and interpretation of results using only mathematically verified output allows for all outputs to be based on deterministic financial logic with the ability to provide human-readable strategic insights.

The elements of systematic generation of scenarios, constraints-based filtering, multi-objective optimization, and AI-assisted interpretation work together as one framework, which does not exist in any current M&A decision-support software.

10. Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence): All testing details

Yes, the system has been implemented existence and undergone testing in a controlled computer-based test setting. The main portion of the system developed, which includes all the essential features contained within the Financial Computation Model, scenarios generation using Grid Search, Verification through the Application of Constraints, and Pareto/Frontier extraction through Programmatic Implementation.

The scenarios that were validated programmatically exhibited varying amounts of financial input that would produce different results based on the way that each simulated deal (Merger and Acquisition) was structured, including multiple values, Synergy Assumptions, and Financial Structures considered when implementing the results of these tests. To ensure that the EPS calculations made after the deal closed, constraints being filtered were accurate, and the identification of Pareto-efficient configurations was correct; testing against the above scenarios was provided to ensure that the results would yield consistent, valid, and meaningful returns on risk based on the input provided by the tester.

The system was tested as an Academic Project in which the operation of the system occurred within a Local Computing Environment. The results of all operations exhibited consistent generation of valid scenarios, and there were also useful trade-offs between Return and Risk.

The AI-based interpretation layer was integrated with structured input-output formatting. Testing was performed to ensure that the explanations generated by this layer would be strictly based upon the computation of the results without changing the underlying Financial Logic.

11. Briefly state when and how you first conceived this idea?

The idea for this invention was first conceived during the process of studying financial modeling and mergers and acquisitions (M&A) analysis as part of theoretical coursework. While working with traditional approaches such as accretion/dilution analysis and scenario-based evaluation, it was observed that these methods typically focus on a limited number of manually defined cases and do not systematically explore all possible deal configurations. This led to the realization that M&A deal structuring could be approached as an optimization problem, where multiple combinations of financing structures and synergy assumptions could be evaluated simultaneously under defined constraints. The concept evolved further by incorporating the idea of balancing return and risk through measurable parameters such as post deal EPS and sensitivity to synergy variations. The idea was increasingly refined during the development of the project, leading to the formulation of a structured system that integrates scenario generation, constraint validation, and multi-objective optimization for decision support.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

The invention has not been sold, offered for sale, or publicly disclosed in any non-confidential manner. The details of the invention have only been shared within an academic environment, such as discussions with faculty members, for the purpose of project development and evaluation. No formal publication, public presentation, or commercial disclosure has been made. To the best of the inventor's knowledge, no non-disclosure agreements were formally executed; however, the disclosure remained limited to an academic and non-public context.

13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

The proposed invention would not have been obvious to a person having ordinary skill in the field of financial modeling or M&A analysis. Existing approaches in this domain chiefly rely on spreadsheet-based evaluation, scenario analysis, and accretion/dilution methods, which are typically used independently and focus on limited, manually selected configurations. The invention introduces a structured transformation of M&A deal analysis into a constraint based multi-objective optimization problem. This shift from static evaluation to systematic exploration of the decision space is not a direct or ordinary extension of active practices. It requires the integration of concepts including grid-based scenario generation, constraint validation, and Pareto efficiency, into a unified computational framework.

Furthermore, the synchronous consideration of return (post-deal EPS) and risk (sensitivity to synergy variations) through formal mathematical modeling represents a non-trivial extension beyond traditional financial analysis, where such trade-offs are not explicitly optimized. Additionally, the design choice of separating deterministic financial computation from AI based interpretation is not an obvious implementation, as most existing systems either rely purely on manual analysis or directly apply predictive models without structured optimization. The invention, therefore, represents more than a direct combination of known techniques, as it introduces a coordinated framework that enables systematic, constraint driven optimization and interpretable decision support in M&A deal structuring.

14. Additional comments by inventor (if you want to give more details out of scope of this IDF).

The proposed system is designed as a structured decision support framework and currently operates under simple financial assumptions to ensure deterministic and reproducible computation. While this allows consistent evaluation of different deal scenarios, real world M&A transactions may involve additional complexities such as tax implications, debt financing structures, and market driven uncertainties, which can be incorporated as extensions to the model. The invention is flexible in nature and can be extended to include additional financial parameters or alternative optimization objectives beyond EPS, depending on special use cases. It can also be adapted for integration into existing financial modeling platforms or deployed as a standalone analytic tool for academic and professional use.

15. Drawings/Flowchart/Table

Fig. 1: Input Interface / Parameter Space
Fig. 2: Grid Search / Scenario Generation Flow
Fig. 3: Financial Model Flow Diagram
Fig. 4: Constraint Validation Process
Fig. 5: Sensitivity / Risk Evaluation Flow
Fig. 6: Pareto Filtering Logic
Fig. 7: Scatter Plot with Pareto Curve
Fig. 8: AI Pipeline Diagram
Fig. 9: Output UI Representation
Fig. 10: Website Dashboard

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